A sample chapter

100 Puzzles

Stories to Solve with Mathematics, Logic and Science

The award book of Math Club & Competitions.

This sample holds the Prologue, where the journey begins, and Puzzle 8, The Swinging Island, with its hint and full solution. Try the puzzle before you turn to the answers.

MATH CLUB & COMPETITIONS



PROLOGUE

The Silent Places

Snow had been falling on Highmere for three days when Master Orla called her four apprentices to the Hall of Maps. The Great Map filled the whole north wall: mountains in brown ink, rivers in blue, roads in fine red lines, a name beside every town. Copies of it hung in every inn and ferry house in the land. Travellers planned their journeys by it, healers found their way to the sick by it, and merchants sent their wagons along its roads.

But here and there the ink had faded, until only a pale square was left. There were a hundred of them.

"Every one of those places is still there," said Orla, tapping a square with her cane. "A harbour, a market, a village on a cliff. People live there as they always have. But each of them has fallen silent. A gate that will not open, a light that will not burn, a quarrel nobody can settle: one trouble, and the wagons stop going, the letters stop coming, and the road grows over. A road nobody travels cannot stay on my map, and the place fades with it."

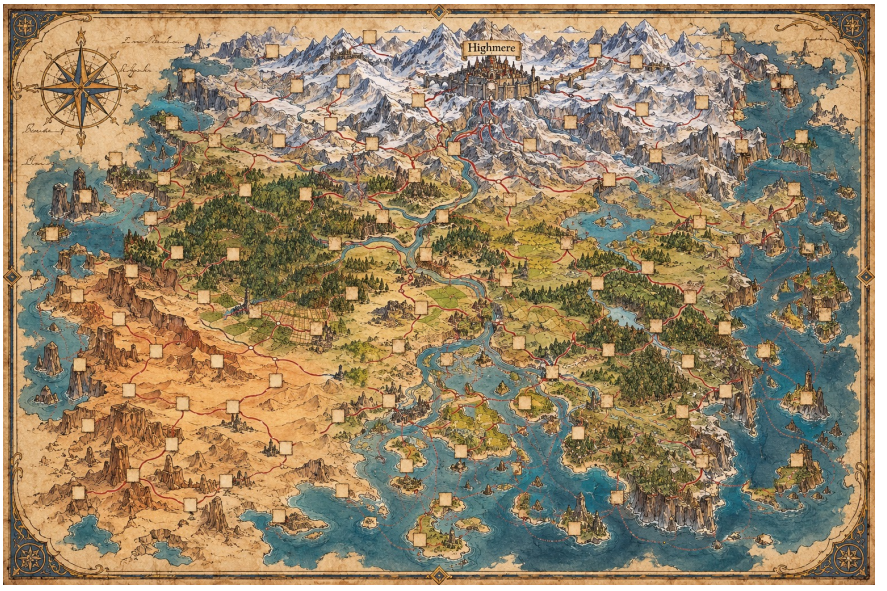
"Then we go and fix them," said Tavi.

"You go and *help*," said Orla. "Mend each trouble, truly and carefully, and people will travel there again. Then Mira can ink its roads back in, and the place is part of the world once more."

She gave them until next midwinter, when they would meet again at the mountain gate. "Not alone, and not all together: you will travel by three roads, and change companions as the seasons change. Mira, the mapmaker, carries the fresh sheets and redraws each place. Whoever is on another road sends Mira a sketch in a letter, and Quill will carry the letters." On the windowsill, a raven with one silver feather shook out her wings.

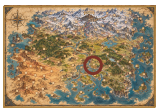
Orla's voice grew quieter. "One more thing. Long ago, another apprentice of mine set out to find these places, and never came home. If you ever find signs of someone who went before you, write them down. I would like to know."

That night the four of them set out under falling snow: three lanterns going three different ways down the mountain, and one raven flying between them.



The Great Map of Highmere. The pale squares are the hundred places that have fallen silent.





PUZZLE 8

The Swinging Island

Snow was coming. Suri and Bao felt it in the wind as the floating monastery of Skytether creaked on its chain, high above a white valley of farms and frozen streams. The whole island was moored by one iron chain, fixed under the bell tower, and the winter wind pushed it slowly round that chain like a weathervane.

Sister Lumen, who kept the monastery's stores, spread an old survey map of the valley across the table. "Six villages are waiting for their winter baskets," she said. "Tonight the wind will turn us once all the way round before the storm. We lower baskets from the landing gate, and the gate goes wherever the island takes it."

The map was ruled in equal squares, and every spot on it was written as two numbers: squares across from the corner, then squares up. The spot below the bell tower was $(8, 8)$, and the spot below the gate $(13, 13)$.

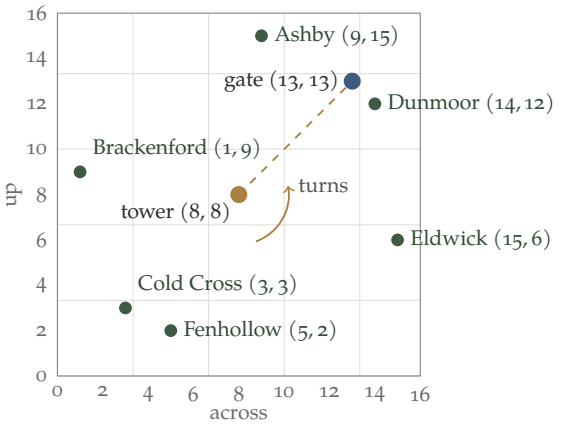
Suri traced a circle with her finger. "Anything that turns about one point keeps the same distance from it, like the handle of a door swinging round its hinges."

"Here is everything we may do," said Sister Lumen.

1. Lower a basket whenever the gate is exactly above a village. A basket lowered anywhere else is lost in the snow.
2. The rope hangs straight down; a basket cannot be swung sideways.
3. The island turns one full circle, slowly, and cannot be steered, stopped or moved off its chain.

"Easy," said Bao, already packing one for Dunmoor. "We just wait for the gate to reach each village. Dunmoor is practically next to it!"

"If the gate ever gets there," said Sister Lumen. "It only goes where the island carries it." Above them, the chain creaked again.



The island turns round the tower spot; the gate turns with it.

WHAT YOU NEED TO KNOW

- If one spot is a squares across and b squares up or down from another, the distance between them, squared, is $a^2 + b^2$ (Pythagoras).

CAN YOU HELP SURI AND BAO?

1. Which villages will the gate pass over? Is Bao right about Dunmoor?

2. Sister Lumen wants a new store hut where grid lines cross, right under the gate's path. Find every such crossing, and explain why there are no others.

3. Two villages that get baskets are exactly a quarter turn apart around the tower. Which two? How can you tell from the grid, without measuring an angle?

BEFORE YOU READ ON

HINTS

The gate always stays the same distance from the bell tower. Work with distance squared: for the gate it is $5^2 + 5^2$. Which other pairs of squares add up to the same number?

Have you tried all three questions? The solution is overleaf.

SOLUTION 8

The Swinging Island

Tags: Circles; Pythagoras; Sums of squares; Quarter turns

1. The gate's path

The island turns about the spot below the bell tower, $T = (8, 8)$, so the gate always stays the same distance from T: it travels round a circle with T at its centre. The gate starts 5 squares across and 5 squares up from T, so its distance squared is $5^2 + 5^2 = 50$.

A village gets a basket exactly when its distance from T, squared, is also 50. Working with distance squared avoids square roots, and two distances are equal exactly when their squares are.

2. The six villages

Village	Across from T	Up from T	Distance squared	Basket?
Ashby	$+1$	$+7$	$1 + 49 = 50$	yes
Brackenford	-7	$+1$	$49 + 1 = 50$	yes
Cold Cross	-5	-5	$25 + 25 = 50$	yes
Dunmoor	$+6$	$+4$	$36 + 16 = 52$	no
Eldwick	$+7$	-2	$49 + 4 = 53$	no
Fenhollow	-3	-6	$9 + 36 = 45$	no

Dunmoor is only a little too far out: the gate passes about a seventh of a square inside it. But the rope hangs straight down, so "nearly" is not enough.

> Ashby, Brackenford and Cold Cross get their baskets.
> Bao is wrong: Dunmoor is just outside the gate's circle.

3. Every crossing on the circle

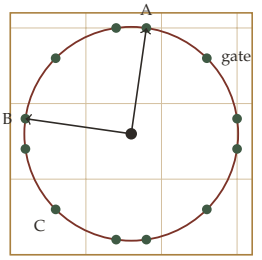
A crossing a squares across and b squares up or down from T is on the circle exactly when $a^2 + b^2 = 50$. Each of a^2 and b^2 is a square number no bigger than 50: 0, 1, 4, 9, 16, 25, 36 or 49. Subtract each from 50:

$$50,\ 49,\ 46,\ 41,\ 34,\ 25,\ 14,\ 1.$$

Only 49, 25 and 1 are square numbers, so the only ways are $1 + 49$, $25 + 25$ and $49 + 1$. Each step can go either way (left or right, up or down), which gives

$$(\pm 1, \pm 7), \qquad (\pm 5, \pm 5), \qquad (\pm 7, \pm 1)$$

from T: four crossings each, twelve in all, and they all fit on the map. On the map they are $(9, 15)$, $(7, 15)$, $(9, 1)$, $(7, 1)$, $(1, 9)$, $(1, 7)$, $(15, 9)$, $(15, 7)$, $(3, 3)$, $(3, 13)$, $(13, 3)$ and $(13, 13)$.



Twelve crossings, shown as dots in the drawing. One of them, $(13, 13)$, is where the gate starts.

4. A quarter turn without angles

Turn the map a quarter turn anticlockwise about T. A step "1 right" becomes "1 up", and a step "7 up" becomes "7 left". So the journey from T to Ashby, 1 right and 7 up, becomes 7 left and 1 up: exactly the journey from T to Brackenford. The two arrows in the drawing are at right angles.

The same trick splits the twelve crossings into three groups of four, each turned into the next by a quarter turn. Written as steps from T:

$$(5, 5) \to (-5, 5) \to (-5, -5) \to (5, -5),$$
$$(1, 7) \to (-7, 1) \to (-1, -7) \to (7, -1),$$
$$(7, 1) \to (-1, 7) \to (-7, -1) \to (1, -7).$$

The last two groups are different: $(1, 7)$ and $(7, 1)$ are mirror images, not quarter turns of each other. Cold Cross, at $(-5, -5)$, is in the first group; Ashby and Brackenford are in the second, so Cold Cross is not a quarter turn from either of them.

Ashby and Brackenford: a quarter turn changes the step 1 right and 7 up into 7 left and 1 up.

On the map *Sister Lumen let Suri and Bao copy her survey map of Skytether's valley, and Suri added the circle the gate sweeps over the valley. Quill carried both to Mira.*